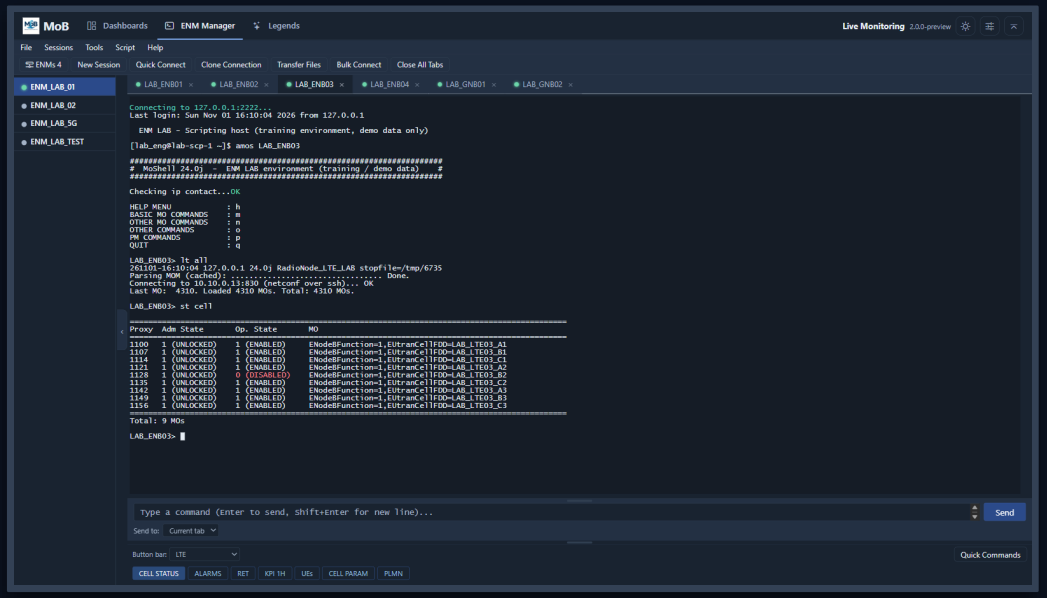
ENM Manager

User guide for RAN engineers

Live Monitoring 2.0 · MoB Tool Box



Every screenshot in this guide uses a LAB environment: demo ENMs, demo sites and demo data.

OVERVIEW

One tool for the daily ENM work

A full SSH client that runs in the browser, built around the tasks a RAN engineer repeats every day.

Multi-channel terminal

One tab per site, with amos, moshell, cmedit and vi working as usual.

Bulk Connect

Opens one channel per site from a .txt list and loads each node by itself.

Quick command buttons

Your commands, one click away, grouped in bars, ready for a list of sites with @sitelist.

ENM CLI capture

Ctrl+Enter runs a cmedit or alarm query and saves it as xlsx, csv or txt.

Transfer Files

Drag files between your PC and the ENM, and get automatic downloads.

Scheduler and PMIC

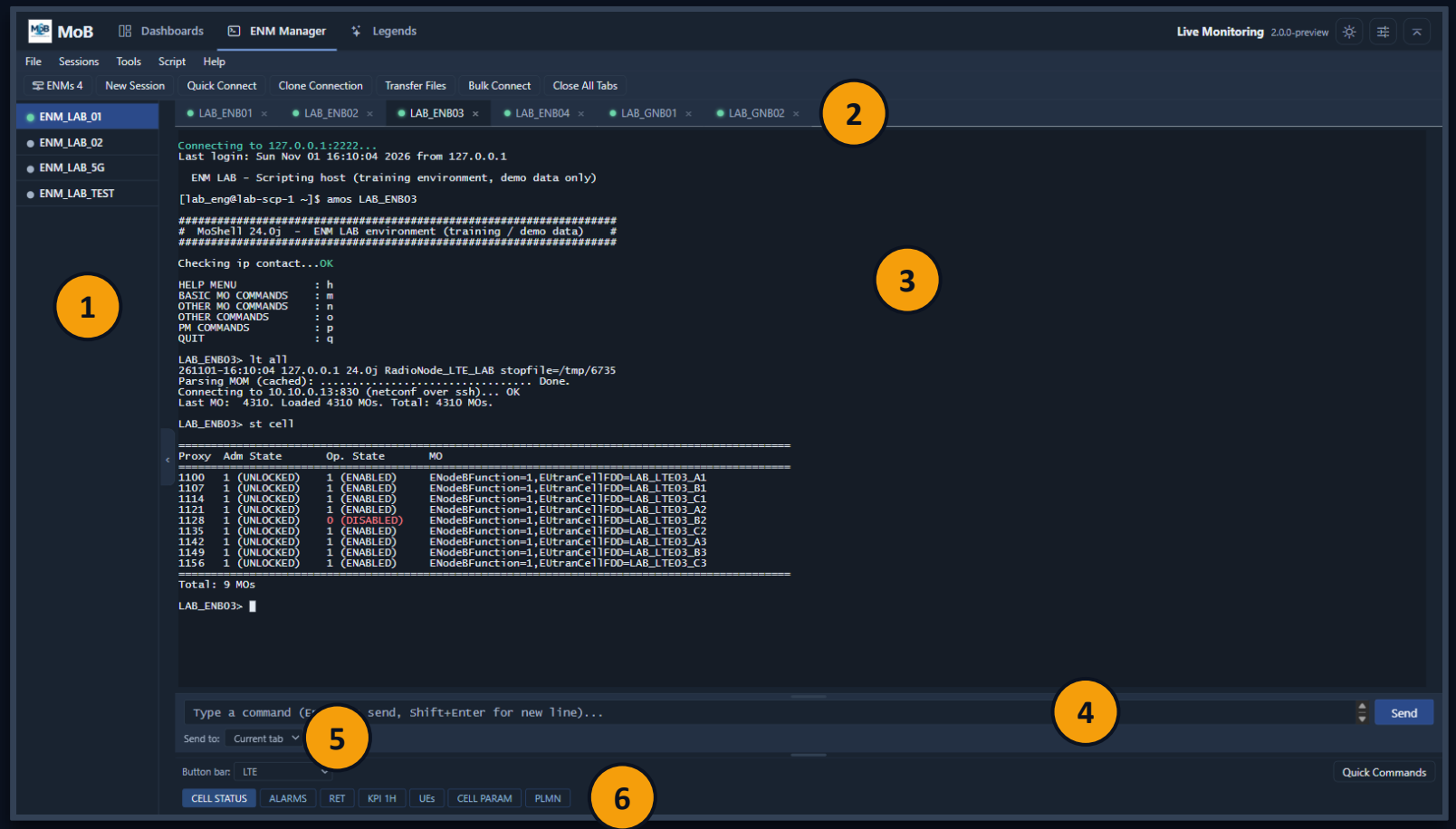
amosbatch and PM counters collected every 15 min, unattended.

01

Getting connected

Sessions, Bulk Connect and moving between channels

The workspace at a glance



1. **ENM list**
   Double-click a session to connect.

2. **Channels**
   One tab per connection, named after the node you are on.

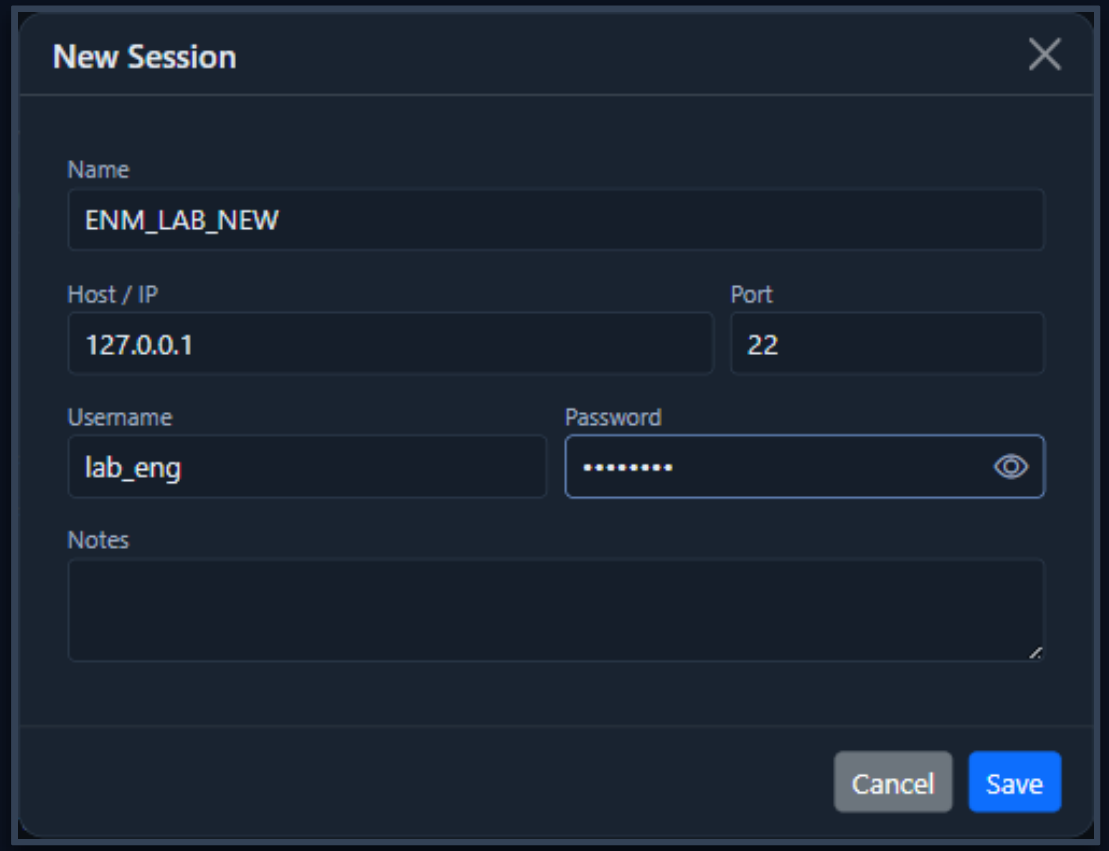
3. **Terminal**
   A real SSH channel: anything you would run on the scripting host works.

4. **Command box**
   Type here: history, multi-line blocks and Ctrl+Enter capture.

5. **Send to**
   Current tab or All tabs.

6. **Button bar**
   Your saved commands, grouped by bar.

Create a session once, then connect with a double-click



1. **Click New Session**
   Give it the name of the ENM. That is the name shown on the tab.

2. **Fill in host, port, user and password**
   Save. The session appears in the ENM list on the left.

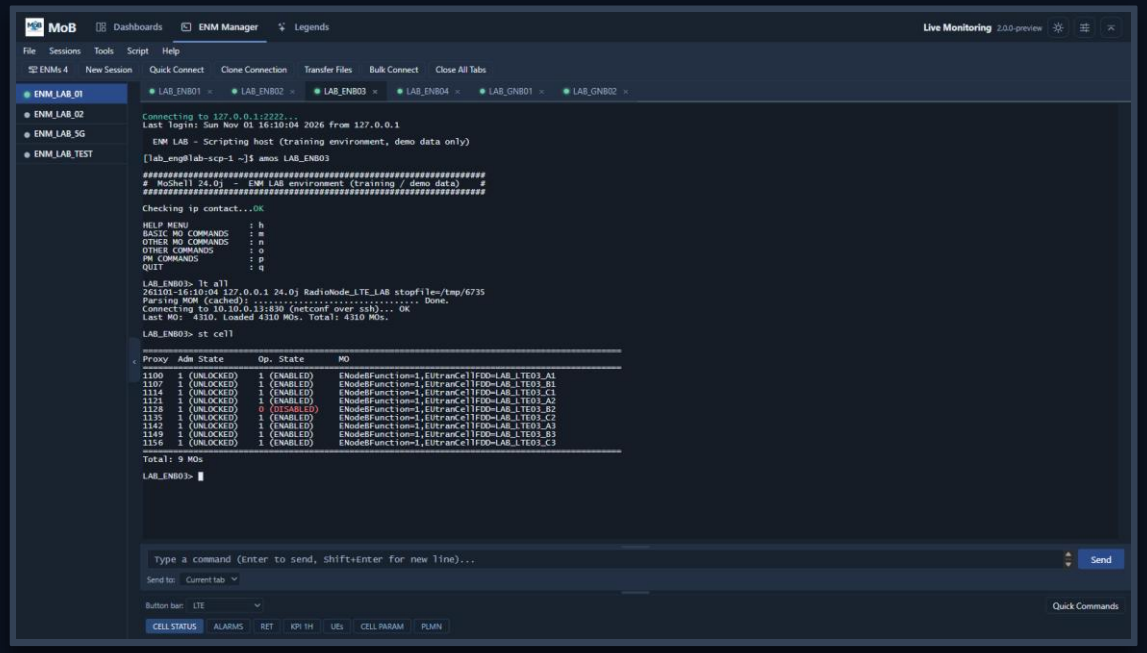
3. **Double-click the session**
   The channel opens. The dot in the list turns green.

4. **Need a one-off host?**
   Quick Connect opens a channel without saving anything.

Saved passwords are protected by your Windows login. They only open for you, on your computer, so you never type a master password.

Bulk Connect: one channel per site





1 **Prepare a .txt**

One site per line. Blank lines and lines starting with # are ignored.

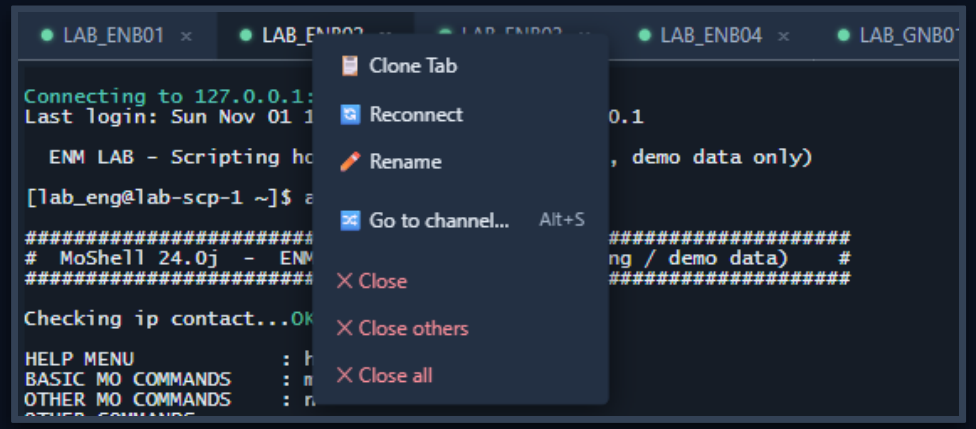
2 **Pick the ENM and the file**

The window lists the sites found. Copy list puts them on the clipboard.

3 **Click Connect**

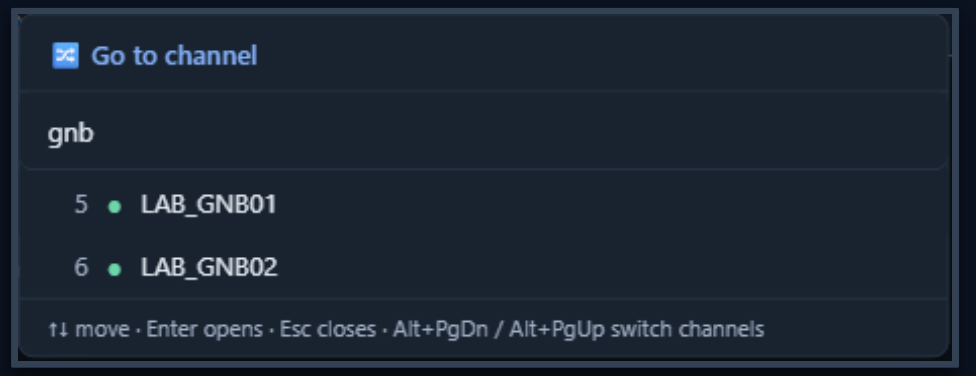
Each tab runs amos <site> and lt all on its own. Your nodes are ready to use.

Move between 20 channels without the mouse



Right-click a tab

Clone, Reconnect (rebuilds the SSH session in the same tab, keeping the history), Rename and Close.



`Alt + PgDn`	Next channel (wraps around)
`Alt + PgUp`	Previous channel
`Alt + S`	Go to channel: type part of the site name or its position
`Double-click tab`	Clone the channel on the same ENM
`Drag a tab`	Reorder the tab bar
`Ctrl + wheel`	Terminal font size (8 to 24 pt)
`Ctrl + End / Home`	Jump to the latest output or to the top
`Alt + wheel`	Scroll 10 times faster

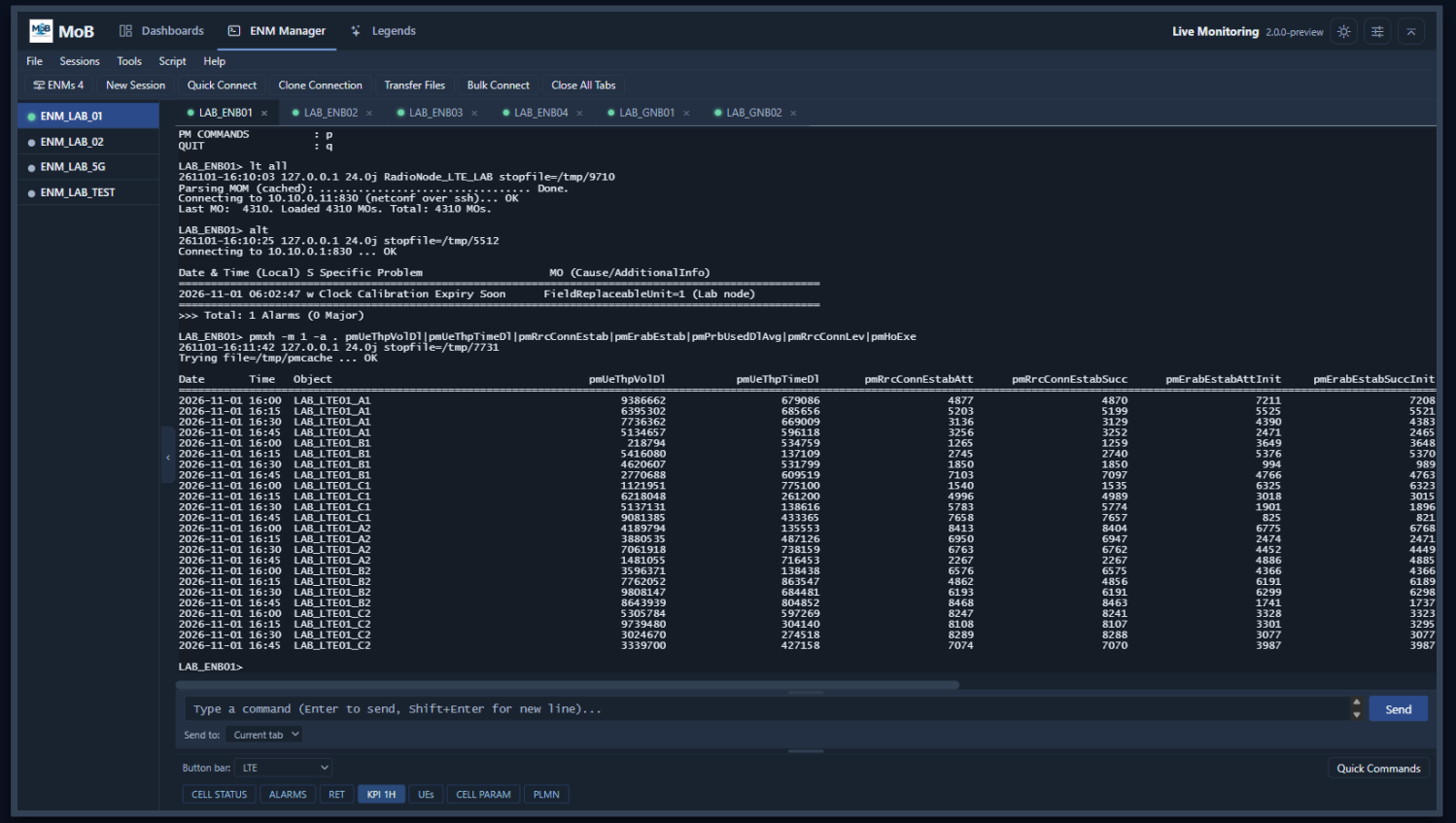
A channel stopped answering after lunch? That is the idle timeout. Right-click the tab and choose Reconnect.

02

Sending commands

The command box, wide outputs and broadcast with protection

The command box does more than Enter



Wide moshell tables are never folded: drag the horizontal bar to read the rest.

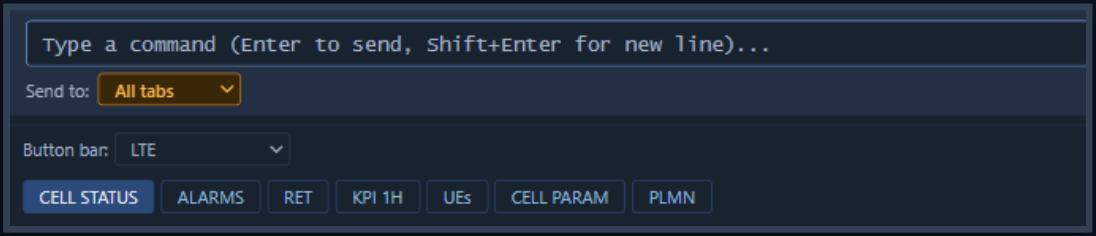
`Enter`	Send
`Shift + Enter`	New line, no send
`↑ / ↓`	Command history
`Ctrl + Enter`	Capture and save
`Ctrl + C`	Copy, or interrupt

Multi-line blocks

Write one command per line and press Enter. The lines go in sequence, with a counter (Sending 3/12) and a Cancel button.

```
lt all
alt
st cell
```

Broadcast to all channels, with a safety net



Send to: All tabs turns ORANGE

One Enter reaches every open channel, and each channel is usually a different node. The color warns you before you press Enter.

BLOCK

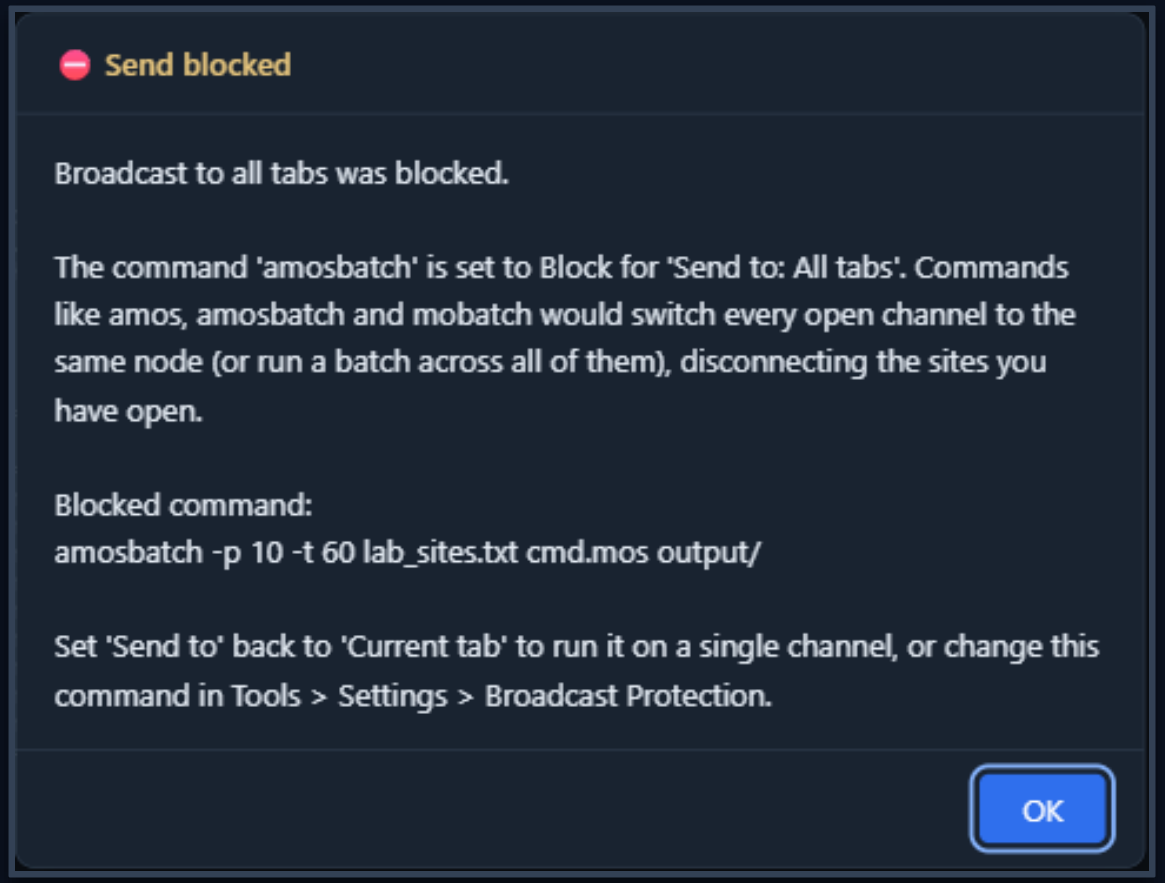
`amos` · `amosbatch` · `mobatch`

Refused: they would move every channel to the same node.

WARN

`set` · `setm` · `lset` · `del` · `rdel` · `acc` · `bl` · `bls`

Asks for confirmation and shows the number of channels.



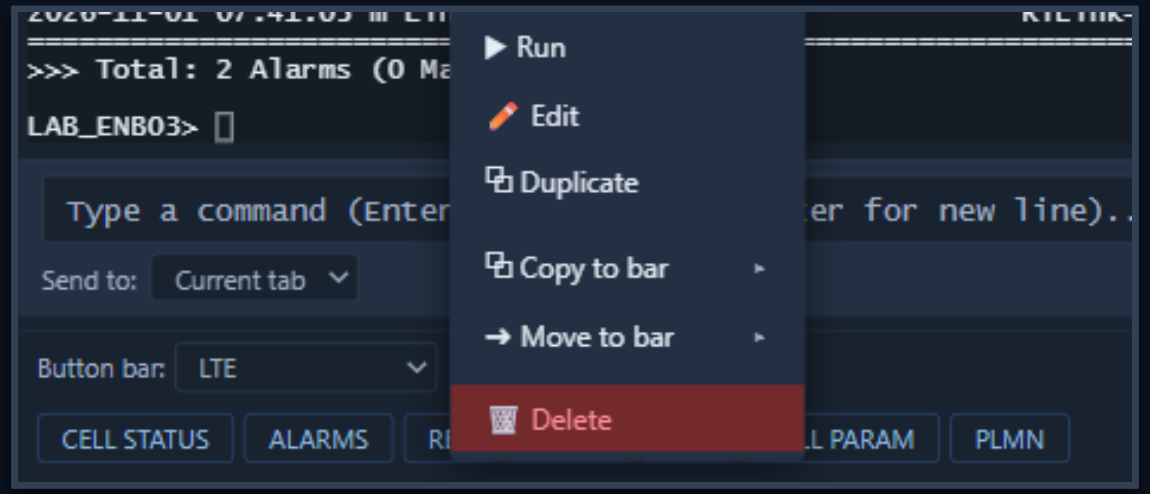
Every line of a multi-line block is checked, and so are the buttons. Edit the list in Tools › Settings.

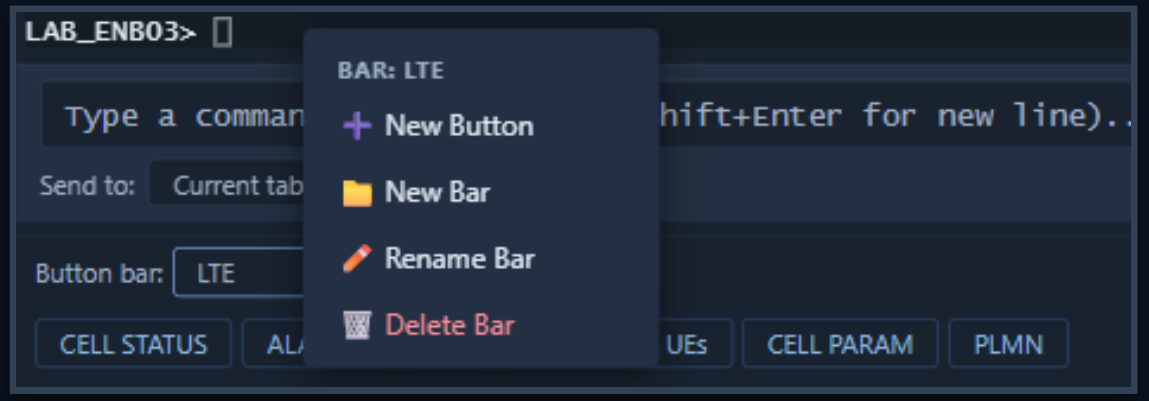
03

Quick command buttons

Save your commands once and run them with one click

Buttons live in bars. Everything is a right-click away





Right-click a BUTTON

Run · Edit · Duplicate · Copy to bar · Move to bar · Delete

Right-click the BAR row

New Button · New Bar · Rename Bar · Delete Bar. The menu names the bar it acts on.

Group by activity

LTE, NR, CLI, Alarms… switch with the Button bar selector.

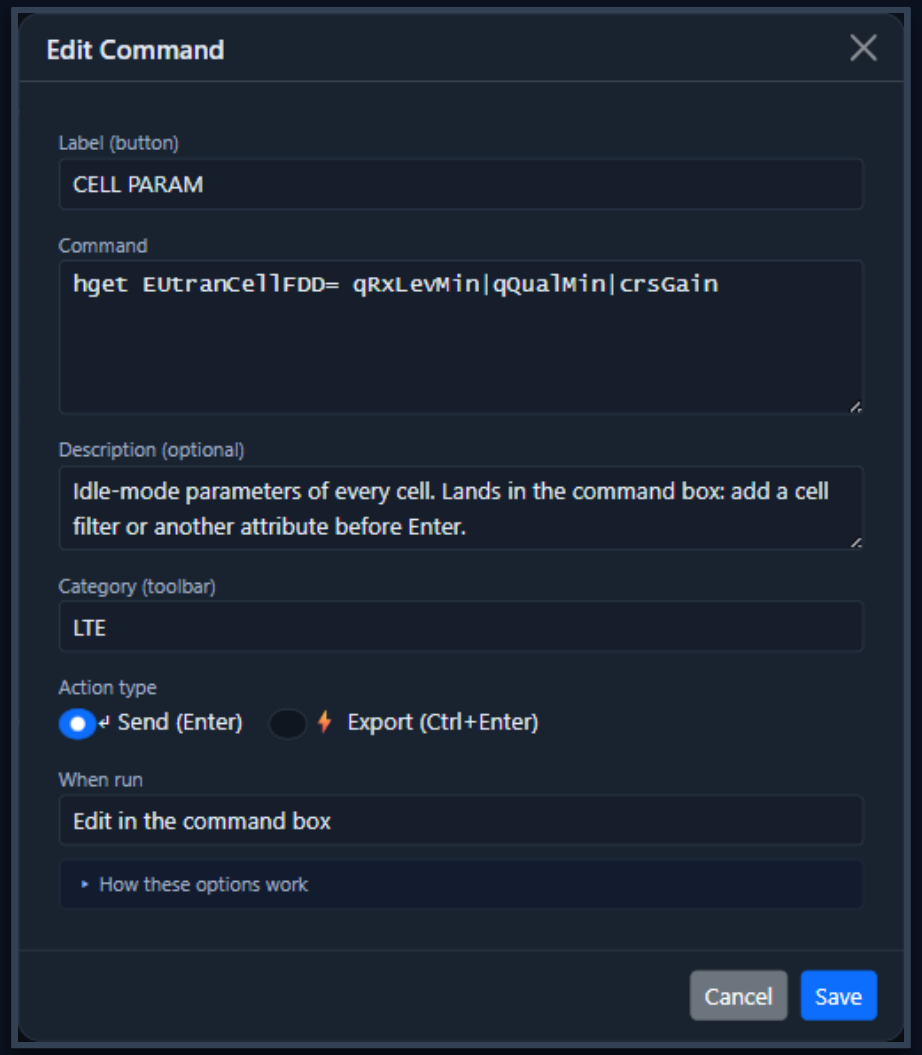
The ⚡ icon

Marks an Export button: it captures the output instead of printing it.

Several lines

A button can hold a whole block, sent line by line.

Decide what a click does



Type of action

Send: types the command in the terminal, like Enter.

Export: captures the output and opens the save window, like Ctrl+Enter.

When run

Run immediately: no questions asked.

Preview & edit before sending: shows the final command, editable, before it leaves. Best with @sitelist.

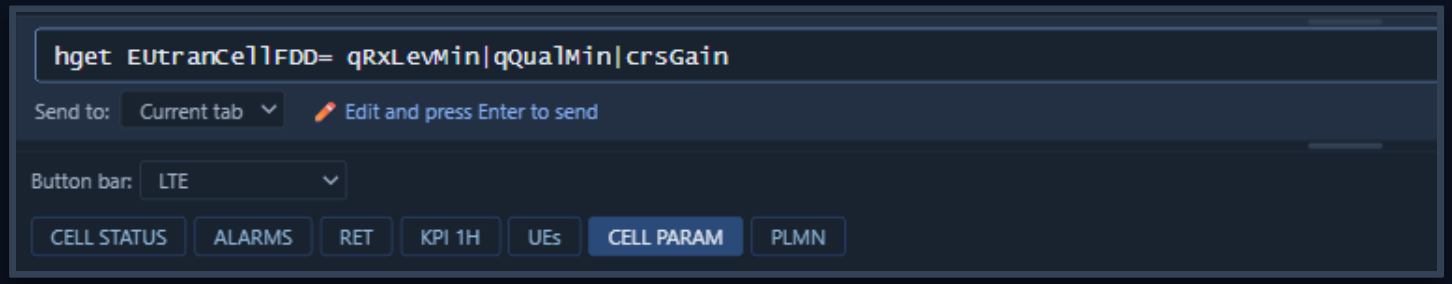
Edit in the command box: drops it in the box and waits for your Enter. It stays in the history, so ↑ brings it back for the next site.

Description

Free text explaining what the button is for. It shows first when you hover the button, which helps a colleague who receives it.

Edit in the command box: adjust, then Enter

Set When run to Edit in the command box. A click drops the command in the box and waits: change what you need and press Enter.



The command waits in the box with the hint Edit and press Enter to send.

Typical uses

`hget EUtranCellFDD= …` → add `=LAB_LTE03_B2` to see one cell

`pmxh -m 1 …` → change to `-m 4` for the last 4 hours

Works with multi-line buttons and with Send to: All tabs.

1. **Click the button**
   The command lands in the box, nothing is sent yet.
2. **Adjust it**
   Narrow the filter, add an attribute, change a counter or the number of hours.
3. **Press Enter**
   It is sent and stays in the history: ↑ brings it back.

Choose the scope: your cluster or the whole network

Whole network: one MO from every node

```
cmedit get * TermPointToGNodeB.(…) -t
```

Only your cluster: sites from a .txt

```
cmedit get @sitelist TermPointToGNodeB.(…) -t
```



1. **Point the button to a .txt**
   One site per line, # for comments. Wildcards like *_PA pass through.
2. **Choose the separator**
   @sitelist joins with ; and @sitelist, joins with a comma.
3. **Pick the scope per button**
   @sitelist for a cluster or an event. * when you need the MO for the whole network.

With When run = Preview, you see the expanded list, and can edit it, before anything is sent.

Find any button in seconds



1. **Tools › Quick Commands**
   Every button you have, from all bars.

2. **Search inside everything**
   Label, command text, description and bar name, with the matches highlighted.

3. **Edit or delete in place**
   Or create a new button with + New Command.

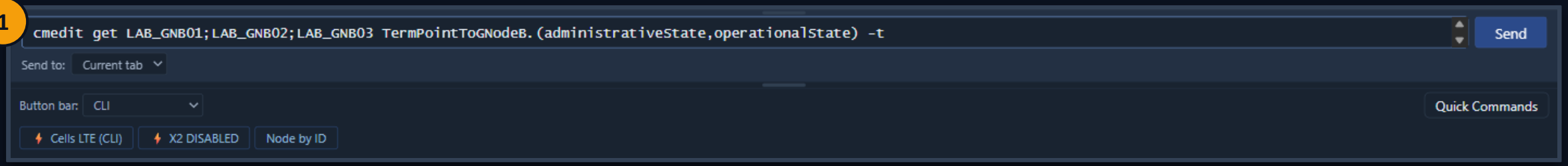
Useful when you remember the counter (pmPdcpVol…) but not the button name or its bar.

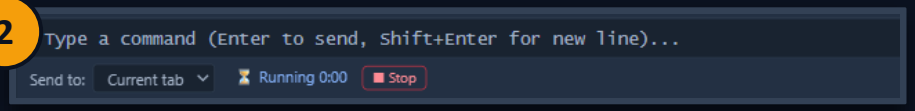
04

ENM CLI and exports

Capture cmedit and alarm queries straight to Excel

Ctrl+Enter: capture the result instead of scrolling

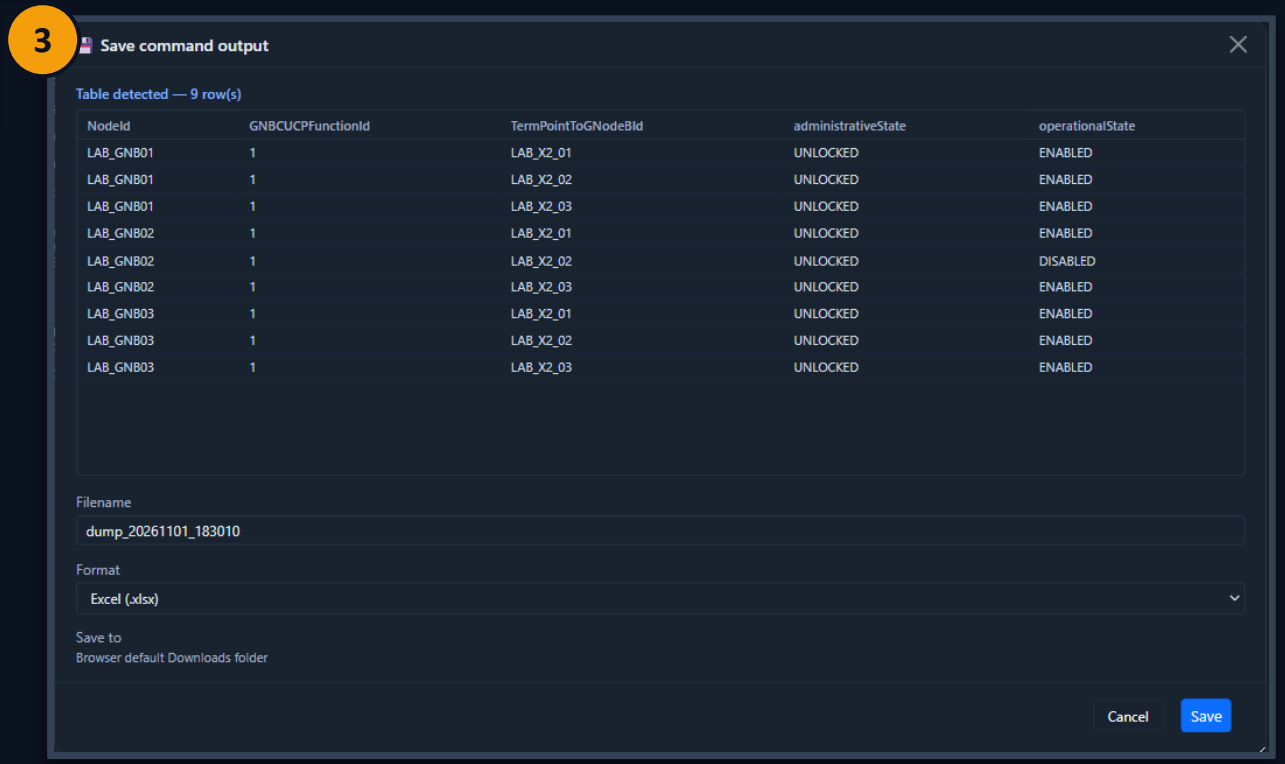




Running 0:42 · 2.1 MB received [Stop]. Time and bytes show the query is alive. A second Ctrl+Enter is refused while one runs.

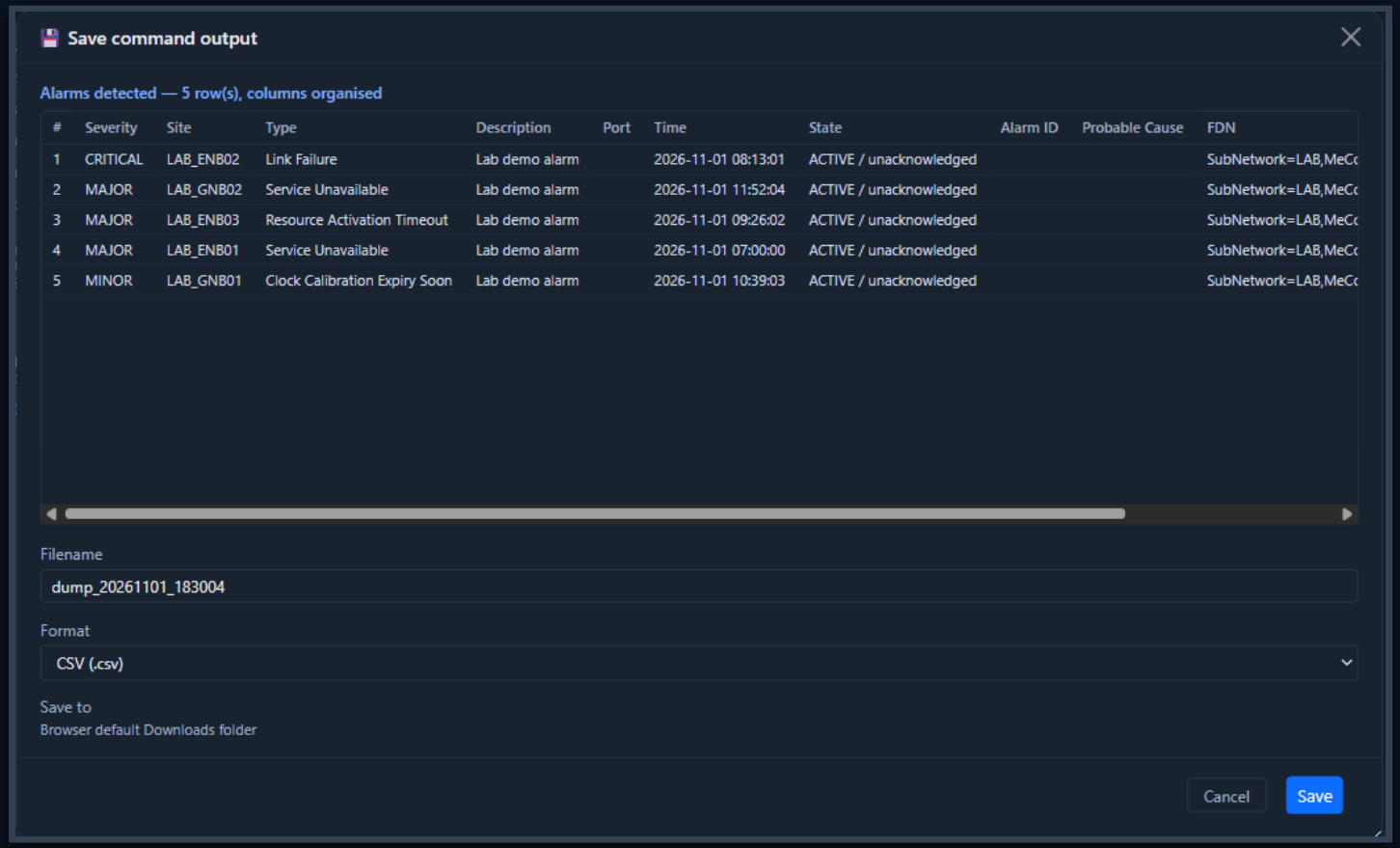
Review the table, then save it as .xlsx, .csv or .txt.

Prompts, colors and echo are removed. Ctrl+Enter always runs on the current tab, on a separate connection.



Export buttons: CLI results as ready-made tables

Built for ENM CLI. One click runs the query on your cluster (@sitelist) or on the whole network (*) and brings back alarms, parameters or inventory already split into columns, ready for Excel.





1 Action type = Export

The button gets the ⚡ icon. A click captures instead of printing.

2 Choose the scope

@sitelist for your cluster, or * to pull an MO from the whole network.

3 Review and save

Columns on screen, then .xlsx, .csv or .txt.

Comes out structured

Alarms: alarm get

Parameters: cmedit get … -t

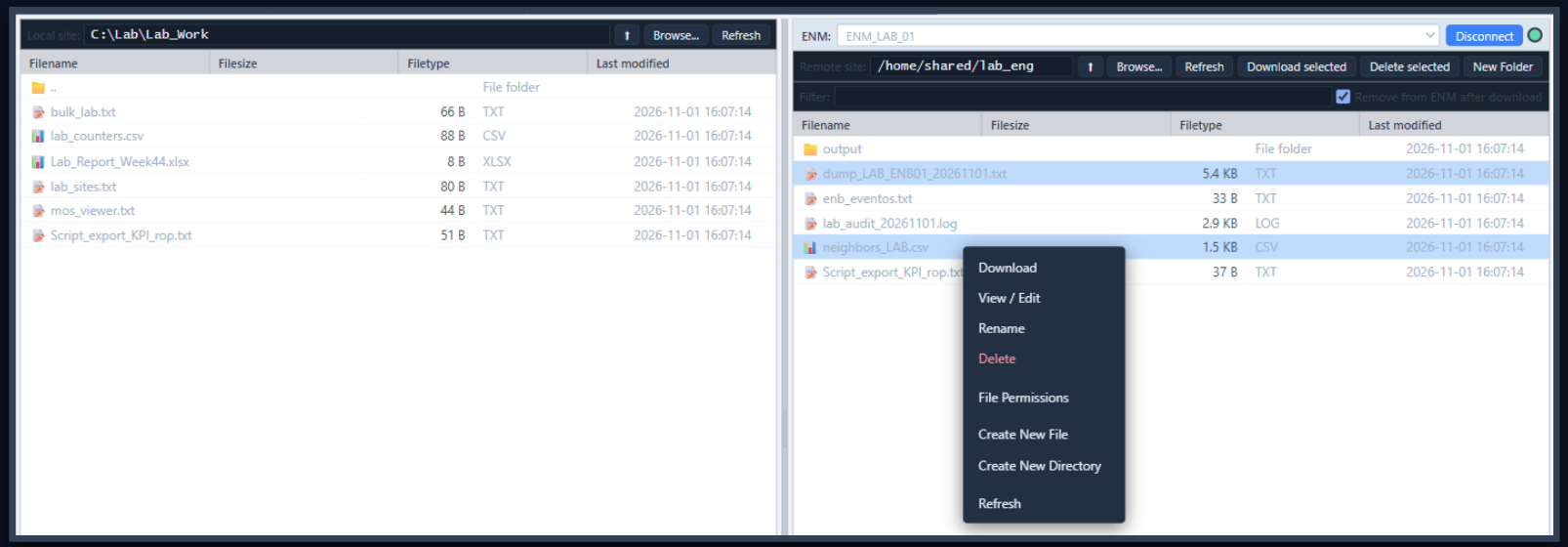
Inventory and relations: cmedit get … -t

05

Files and automation

Transfer Files, Script Scheduler and the PMIC collector

Transfer Files: your PC on the left, the ENM on the right



`Right-click`	Menu for that row
`Ctrl + A`	Select visible rows
`F2`	Rename (PC side)
`Delete`	Delete (asks first)
`Type a letter`	Jump to the name
`Click session`	Switch the ENM

Drag and drop

Left → right uploads, right → left downloads. You can also drop files from Windows.

Filter

Type part of the names: only matching rows are shown and acted on.

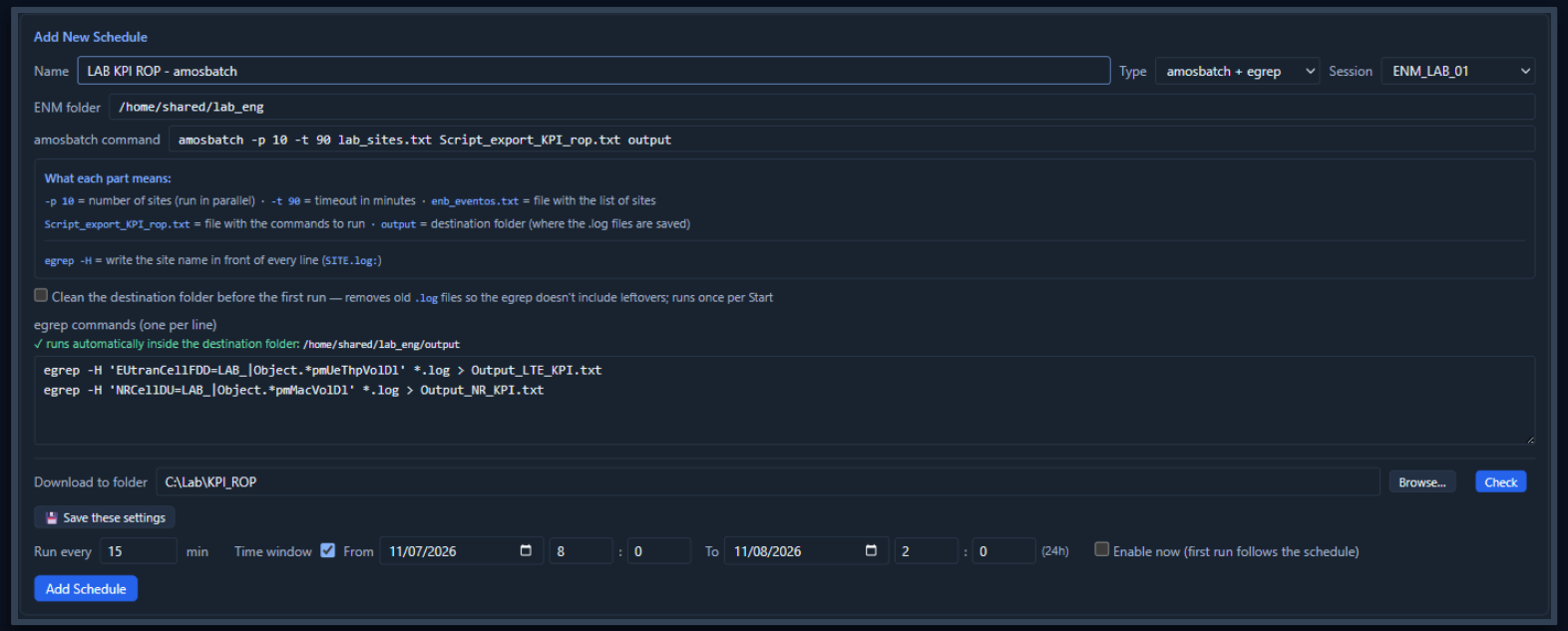
Remove from ENM after download

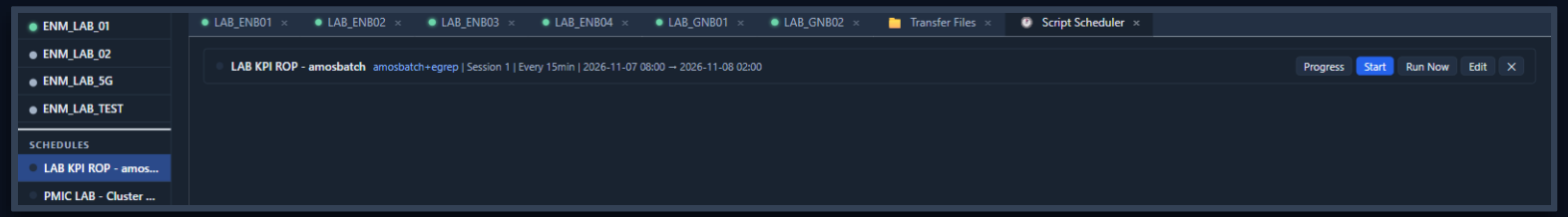
Keeps your ENM home folder clean. System folders are never touched.

Automatic download

When a command prints File saved: /path, the file comes to your PC by itself.

Script Scheduler: amosbatch + egrep on autopilot





1 **Script › Script Scheduler**

Choose the type amosbatch + egrep and the ENM session.

2 **Type the amosbatch command**

-p sites in parallel, -t timeout, site list, command file, output folder.

3 **Add the egrep lines**

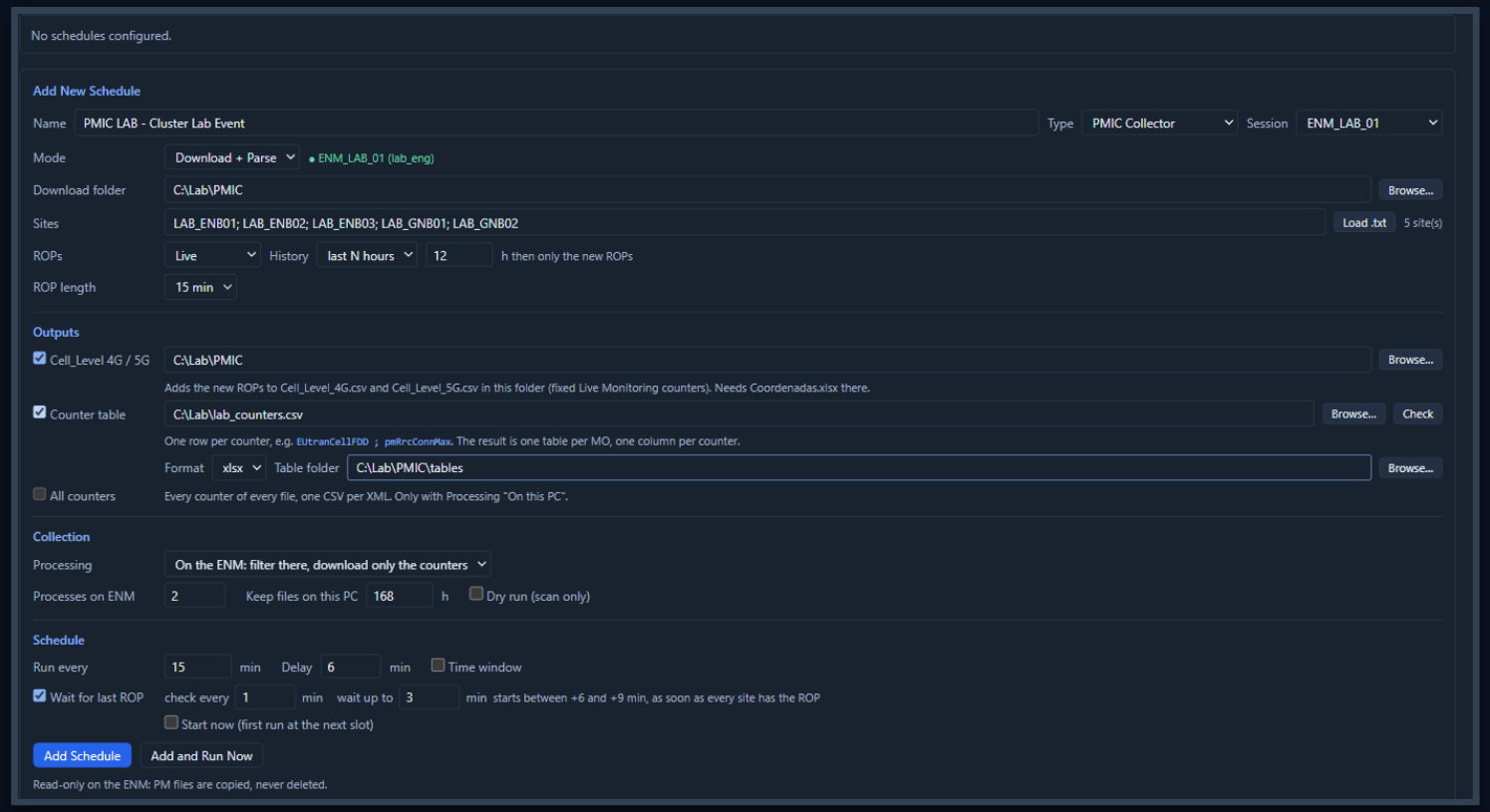
They run in the output folder, and the result is downloaded to your PC.

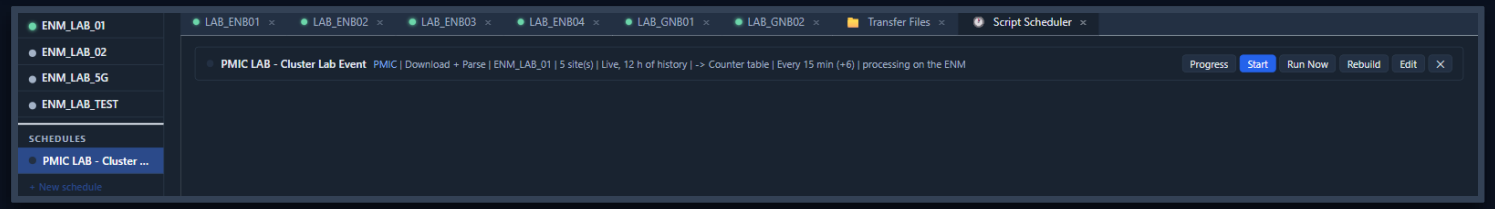
4 **Set interval and time window**

Runs stop by themselves at the end of the event.

Progress shows the live log. A run still going at the next cycle makes that cycle skip, never queue.

PMIC Collector: PM counters straight from the ENM





1 Sites + counter list

Type the sites or load a .txt. The counters come from a .csv or .xlsx with the columns MO and Counter.

2 Outputs

A counter table (one row per object and ROP), the Live Monitoring Cell_Level CSVs, or every counter.

3 Processing on the ENM

The filter runs on the ENM and only a small CSV comes down. It is fast.

4 Wait for last ROP

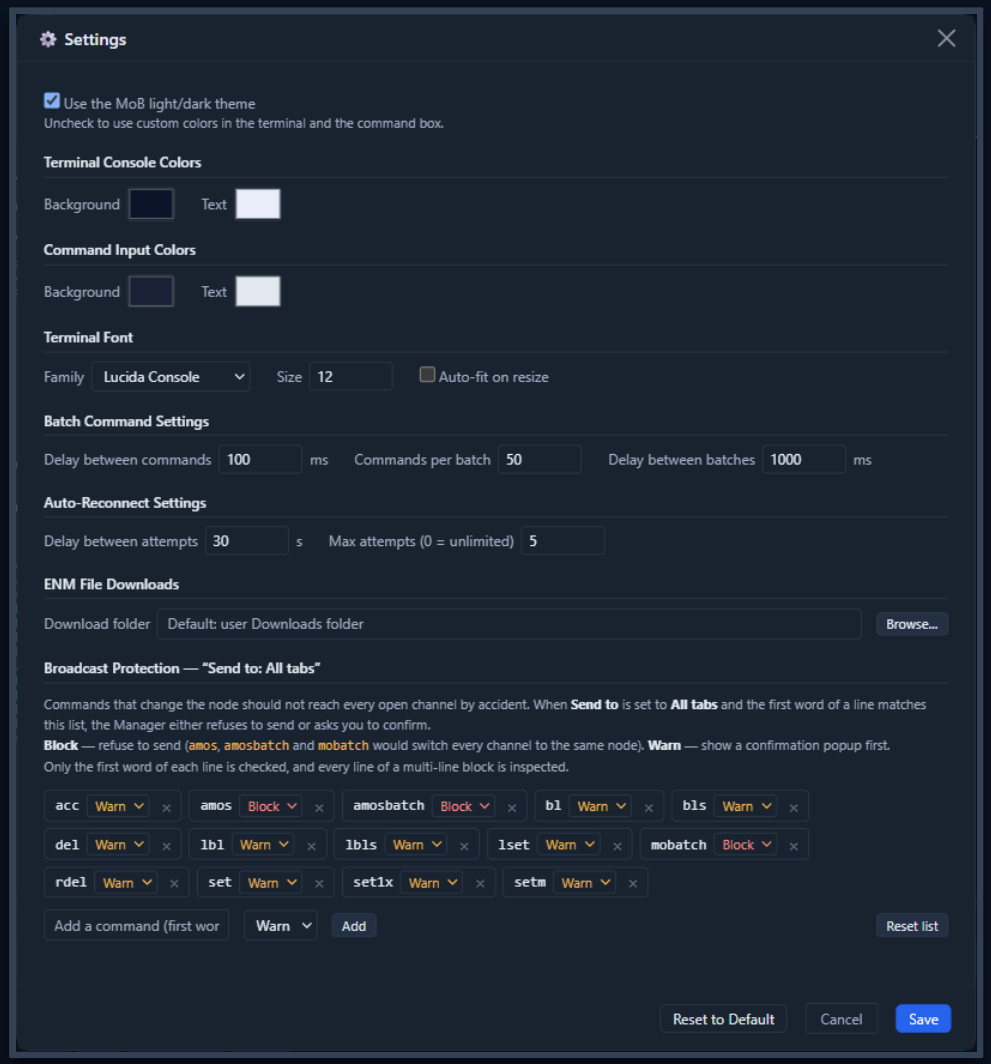
Starts as soon as every site has the ROP. A site that is down never holds the others.

06

Personalise and share

Settings, broadcast protection and sharing with your team

Settings: make the tool yours



Colors and font

Follow the MoB theme or choose your own colors (next slide), plus font family and size.

Batch settings

Delay between lines and batch size for long multi-line blocks.

Auto-reconnect

Delay and maximum attempts when a channel drops.

Download folder

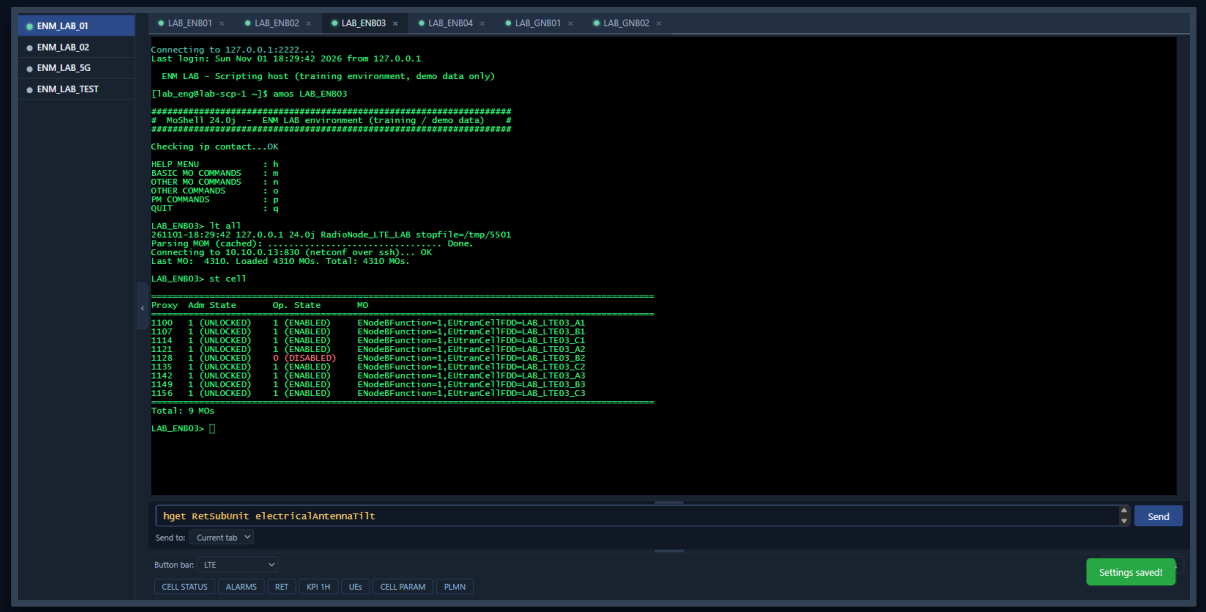
Where automatic downloads and captures land.

Broadcast Protection

Add or remove commands, choose Warn or Block, or reset to the default list.

Your own colors for the terminal and the command box





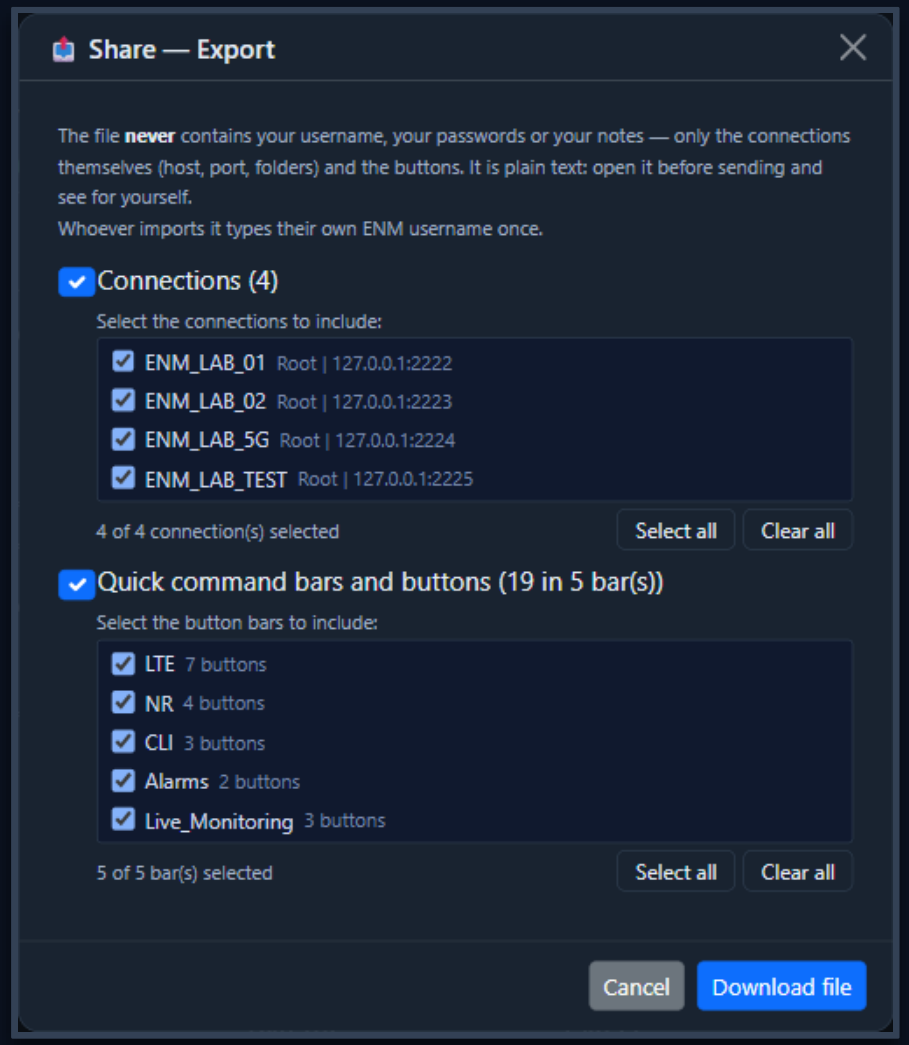
1. **Tools › Settings**
   Untick Use the MoB light/dark theme. The color pickers unlock.

2. **Pick four colors**
   Terminal background and text, command box background and text.

3. **Save**
   Applies at once to every open channel and is remembered on this PC.

Example: classic black terminal with green text and a yellow command box. Font family and size are in the same window.

Share your ENMs and buttons with a colleague



1. **Tools › Export Connections & Buttons**
   Tick only what matters: the two ENMs and the one bar of buttons your colleague needs.

2. **Send the file**
   It is readable JSON. Open it first if you want to see what is inside.

3. **Your colleague imports it**
   Tools › Import asks for their own ENM username. Items that already exist are skipped.

Never included: usernames, passwords, notes or local @sitelist paths. Each person types their own password at the first connection.

Cheat sheet

Keyboard

`Alt + PgDn / PgUp`	Next / previous channel
`Alt + S`	Go to channel
`Ctrl + Enter`	Capture output to xlsx/csv/txt
`Shift + Enter`	New line without sending
`↑ / ↓`	Command history
`Ctrl + wheel`	Font size
`Ctrl + End`	Back to the latest output

Tokens and gestures

`@sitelist`	Sites from .txt, joined with ;
`@sitelist,`	Sites from .txt, joined with ,
`Edit in box`	Button lands in the box: adjust, then Enter
`Double-click tab`	Clone the channel
`Right-click tab`	Reconnect without losing history
`Orange Send to`	You are about to broadcast



Help › Documentation opens the full user guide, the shortcuts page and the technical reference, all with a search box.

Work faster.

Stay safe.

Four habits that get the most out of the ENM Manager

Start the day with Bulk Connect

Your cluster loaded in one click.

Turn repeated commands into buttons

Add a description so the team understands them.

Check the color before Enter

Orange means every channel receives it.

Let the Scheduler collect for you

amosbatch and PMIC run while you analyse.